

8.9

Mycology

1. **All of the following are examples of appropriate specimens for the recovery of fungi except:**

- A. Tissue biopsy
- B. Cerebrospinal fluid
- C. Aspirate of exudate
- D. Swab

Microbiology/Apply knowledge to identify sources of error/Mycology/1

2. **For which clinical specimens is the KOH direct mount technique for examination of fungal elements used?**

- A. Skin
- B. Cerebrospinal fluid
- C. Blood
- D. Bone marrow

Microbiology/Apply principles of basic laboratory procedures/Mycology/1

3. **The India ink stain is used as a presumptive test for the presence of which organism?**

- A. *Aspergillus niger* in blood
- B. *Cryptococcus neoformans* in CSF
- C. *Histoplasma capsulatum* in CSF
- D. *Candida albicans* in blood or body fluids

Microbiology/Correlate clinical and laboratory data/Mycology/2

4. **Cutaneous disease involving skin, hair, and nails usually indicates an infection with a:**

- A. Dimorphic fungus
- B. Dermatophyte
- C. Mucorales (previously Zygomycetes)
- D. *Candida* species

Microbiology/Correlate clinical and laboratory data/Mycology/2

Answers to Questions 1–4

1. **D** Specimens for fungal culture must be kept in a moist, sterile environment. Swabs that are dried out or submitted with insufficient material on them should be rejected. Generally, swabs are inadequate for the recovery of fungi because they are easily contaminated by surrounding skin flora.
2. **A** A solution of 10% KOH is used for contaminated specimens, such as skin, nail scrapings, hair, and sputum, to clear away background debris that may resemble fungal elements. Normally sterile specimens (CSF, blood, and bone marrow) do not require KOH for clearing.

3. **B** Meningitis caused by *C. neoformans* is diagnosed through culture, biochemical reactions, and rapid agglutination tests for cryptococcal antigen. The India ink test is not diagnostic for cryptococcal meningitis because positive staining results are demonstrated in less than 50% of confirmed cases. A positive India ink test shows yeast cells in CSF with a surrounding clear area (the capsule) because the capsule of *C. neoformans* is not penetrated by ink particles.
4. **B** Superficial dermatophytes rarely invade the deeper tissues and are the cause of most cutaneous fungal infections. Fungal infections of the skin are most often caused by *Microsporum* spp., *Trichophyton* spp., and *Epidermophyton* spp., although *Candida* spp. are sometimes implicated as the cause of nail infections.

5. **What is the first step to be performed in the identification of an unknown yeast isolate?**
- A. Gram staining
 - B. India ink staining**
 - C. Catalase test
 - D. Germ tube test

Microbiology/Select methods/Reagents/Media/Mycology/2

6. **An isolate produced a constriction that was interpreted as a positive germ tube, but *C. albicans* was ruled out when confirmatory tests were performed. Which of the following fungi is the most likely identification?**

- A. *Candida tropicalis*
- B. *Cryptococcus neoformans***
- C. *Candida glabrata*
- D. *Rhodotorula rubra*

Microbiology/Apply knowledge of fundamental biological characteristics/Mycology/2

7. **Cornmeal agar with Tween 80 is used to identify which characteristic of an unknown yeast isolate?**

- A. Hyphae (true and pseudo)
- B. Blastoconidia and arthroconidia**
- C. Chlamydoconidia
- D. All of these options

Microbiology/Apply knowledge of basic laboratory procedures/Mycology/1

8. **Blastoconidia are the beginning of which structures?**

- A. Arthroconidia
- B. Germ tubes**
- C. Pseudohyphae
- D. True hyphae

Microbiology/Apply knowledge of fundamental biological characteristics/Mycology/1

9. **An isolate from CSF growing on cornmeal agar produces the following structures:**

Blastoconidia = + Pseudohyphae = Neg

Chlamydoconidia = Neg Arthroconidia = Neg

Which tests should be performed next?

- A. Birdseed agar and urease
- B. Germ tube and glucose**

- C. India ink and germ tube
- D. All of these options

Microbiology/Select methods/Reagents/Media/Mycology/2

10. Which of the following yeast enzymes is detected using birdseed (niger seed) agar?

- A. Phenol oxidase
- B. Catalase
- C. Urease
- D. Nitrate reductase

Microbiology/Apply knowledge of fundamental biological characteristics/Mycology/2

Answers to Questions 5–10

5. **D** The true germ tube (filamentous extension from a yeast cell) is approximately one-half the width and three to four times the length of the cell with no true hyphae constriction at the point of origin. *C. albicans* produce germ tubes (95%), and a positive test is considered a presumptive identification.
6. **A** *C. tropicalis* forms pseudohyphae that resemble true germ tubes by producing a constriction at the point of origin of the yeast cell. Germ tubes represent true hyphae without constriction, and therefore the test should have been repeated along with carbohydrate tests before making a presumptive identification. The other three species of yeast listed do not form hyphae.
7. **D** Cornmeal agar with Tween 80 (polysorbate) reduces the surface tension and allows for enhanced formation of hyphae, blastospores, and chlamydoconidia.
8. **C** Pseudohyphae are the result of a pinching-off process, blastoconidiation, with the growth of filaments with constrictions. Germ tubes are the beginning of true hyphae (no constrictions). Arthrospores are the result of a breaking-off process of true septate hyphae resulting in square conidia.
9. **A** A yeast that is isolated from CSF and is producing blastospores is most likely to be *C. neoformans*, which is positive for urease and produces brown colonies on birdseed agar.
10. **A** Most isolates of *C. neoformans* produce phenol oxidase when grown on *Guizotia abyssinica* medium (birdseed medium), producing brown to black pigmented colonies. *C. neoformans* is the only *Cryptococcus* species that oxidizes o-diphenol to melanin, which is responsible for the color.

	<i>C. neoformans</i>	<i>R. rubra</i>	<i>T. cutaneum</i>	<i>C. albicans</i>
Urease	+	+	+	Neg
Germ tube	Neg	Neg	Neg	+
Brown and black colonies on niger seed agar	+	Neg	Neg	Neg

11. Which of the following yeasts is characteristically positive for germ tube production?

- A. *Candida tropicalis*
- B. *Candida kefyr* (pseudotropicalis)
- C. *Cryptococcus neoformans*
- D. *Candida albicans*

Microbiology/Apply knowledge of fundamental biological characteristics/Mycology/1

12. Arthroconidia production is used to differentiate which two yeast isolates?

- A. *Candida albicans* and *Candida dubliniensis*
- B. *Trichosporon cutaneum* and *Cryptococcus neoformans*
- C. *Candida albicans* and *Candida tropicalis*
- D. *Saccharomyces cerevisiae* and *Candida glabrata*

Microbiology/Apply knowledge of fundamental biological characteristics/Mycology/2

13. The urease test, niger seed agar test, and the germ tube test are all used for the presumptive identification of:

- A. *Rhodotorula rubra*
- B. *Cryptococcus neoformans*
- C. *Trichosporon cutaneum*
- D. *Candida albicans*

Microbiology/Apply knowledge of fundamental biological characteristics/Mycology/2

14. Which of the following yeasts produces only blastoconidia on cornmeal Tween 80 agar?

- A. *Candida* spp.
- B. *Trichosporon* spp.
- C. *Geotrichum* spp.
- D. *Cryptococcus* spp.

Microbiology/Apply knowledge of fundamental biological characteristics/Mycology/2

Answers to Questions 11–14

- 11. D** *C. albicans* and *Candida dubliniensis*, a variant of *C. albicans*, are the only yeasts that produce germ tubes within 1 to 3 hours of incubation at 37°C. *C. tropicalis* produces pseudohyphae after incubation for 3 hours, which may be mistaken for germ tubes. A careful evaluation of the tube origin for constriction is required to avoid a false-positive interpretation.
- 12. B** *T. cutaneum* and *C. neoformans* are both urease positive, but *T. pullulans* produces arthroconidia, and *C. neoformans* does not. In addition to *Trichosporon* spp., arthroconidia are produced by *Geotrichum* spp.
- 13. B** Germ tube–negative isolates producing dark brown to black colonies on niger seed agar and a positive urease test are presumptive of *C. neoformans*. A positive germ tube test result is a presumptive identification for *C. albicans* as well as for *C. dubliniensis*. See the chart above.
- 14. D** *Cryptococcus* spp. do not form either pseudohyphae or arthroconidia. *Candida* spp. produce blastoconidia or pseudohyphae. *Trichosporon* spp. produce pseudohyphae, blastoconidia, and arthroconidia. See the chart below.

Blastoconidia	Pseudohyphae	Arthroconidia

<i>Cryptococcus</i> spp.	+	Neg	Neg
<i>Candida</i> spp.	+	+	Neg
<i>Trichosporon</i> spp.	+	+	+
<i>Geotrichum</i> spp.	Neg	+	+

15. Ascospores are formed by which yeast isolate?

- A. *Saccharomyces cerevisiae*
- B. *Candida albicans*
- C. *Cryptococcus neoformans*
- D. All of these options

Microbiology/Apply knowledge of fundamental biological characteristics/Mycology/2

16. A germ tube–negative, pink yeast isolate was recovered from the respiratory secretions and urine of a patient with AIDS. Given the following results, what is the *most likely* identification?

CORNMEAL TWEEN 80 AGAR

Blastoconidia = +

Pseudohyphae = Neg

Arthroconidia= Neg

Urease = +

-
- A. *Candida albicans*
 - B. *Rhodotorula* spp.
 - C. *Cryptococcus* spp.
 - D. *Trichosporon* spp.

Microbiology/Evaluate laboratory data to make identifications/Mycology/3

17. Terminal chlamydoconidia production is demonstrated by which *Candida* species?

- A. *Candida glabrata*
- B. *Candida krusei*
- C. *Candida albicans*
- D. *Candida tropicalis*

Microbiology/Apply knowledge of fundamental biological characteristics/Mycology/1

18. Carbohydrate assimilation tests are used for the identification of yeast isolates by inoculating media:

- A. Free of carbohydrates
- B. Free of niger seed
- C. Containing carbohydrates
- D. Containing yeast extract

Microbiology/Apply principles of basic laboratory procedures/Mycology/1

19. Yeast recovered from the urine of a catheterized patient receiving chemotherapy for cancer gave the following results:

CORNMEAL TWEEN 80 AGAR

Germ tube = +

Blastoconidia = +

Pseudohyphae = +

Arthroconidia= Neg

Terminal chlamydoconidia = +

What further testing is necessary?

- A. Carbohydrate assimilation and urease
- B. Urease and niger seed
- C. Nitrate reductase and carbohydrate fermentation
- D. No further testing is needed for identification

Microbiology/Select course of action/Mycology/3

Answers to Questions 15–19

15. **A** Sexual spore production is a characteristic of the *Ascomycotina*, which produce an ascus (saclike structure) after the union of two nuclei. The resulting spore is termed *ascospore*. *S. cerevisiae* produces ascospores when grown on ascospore agar for 10 days at 25°C.
 16. **B** *Rhodotorula* spp. produce pink- to coral-colored colonies on Sabouraud agar and cornmeal agar. It is usually considered a contaminant but is an opportunistic pathogen, and must be identified when found in specimens from immunosuppressed patients.
 17. **C** Cornmeal Tween 80 agar supports the growth of *C. albicans* and the formation of its distinctive thick-walled, usually single, terminal chlamydoconidia. *C. dubliniensis* also produces chlamydoconidia (in pairs, triplets, and clusters) and germ tubes but is rarely implicated in disease.
 18. **A** The yeast isolate is inoculated directly into the molten agar base free of carbohydrates or is poured as a suspension onto a yeast nitrogen agar base plate. Carbohydrate disks are then added to the surface of the agar, and the plates are incubated for 24 to 48 hours at 30°C. Growth around the disk indicates the ability of the yeast to utilize the carbohydrate(s) as a sole source of carbon.
 19. **D** This isolate is *C. albicans*, which also produces some true hyphae along with pseudohyphae. A positive germ tube is a presumptive identification along with the production of blastoconidia, terminal chlamydoconidia, and pseudohyphae.
- 20. A blood agar plate inoculated with sputum from a patient with diabetes mellitus grew very few bacterial flora and a predominance of yeast. Given the following results, what is the most likely identification of the yeast isolate?**

CORNMEAL TWEEN 80 AGAR

Germ tube = Neg	Pseudohyphae = +
Arthroconidia = Neg	Blastoconidia = + (arranged along pseudohyphae)
Chlamydoconidia = Neg	

- A. *Candida tropicalis*
- B. *Candida kefyr* (*pseudotropicalis*)
- C. *Trichosporon cutaneum*
- D. *Geotrichum candidum*

Microbiology/Evaluate laboratory data to make identifications/Mycology/3

- 21. Dimorphic molds are found in infected tissue in which form?**
- A. Mold phase
 - B. Yeast phase

- C. Encapsulated
- D. Latent

Microbiology/Apply knowledge of fundamental biological characteristics/Mycology/1

22. The mycelial form of which dimorphic mold produces thick-walled, rectangular, or barrel-shaped alternate arthroconidia?

- A. *Coccidioides immitis*
- B. *Sporothrix schenckii*
- C. *Histoplasma capsulatum*
- D. *Blastomyces dermatitidis*

Microbiology/Apply knowledge of fundamental biological characteristics/Mycology/2

23. The yeast form of which dimorphic fungus appears as oval or elongated cigar shapes?

- A. *Coccidioides immitis*
- B. *Sporothrix schenckii*
- C. *Histoplasma capsulatum*
- D. *Blastomyces dermatitidis*

Microbiology/Apply knowledge of fundamental biological characteristics/Mycology/2

24. The mycelial form of *H. capsulatum* seen on agar resembles:

- A. *Sepedonium* spp.
- B. *Penicillium* spp.
- C. *Sporothrix* spp.
- D. *Coccidioides* spp.

Microbiology/Apply knowledge of fundamental biological characteristics/Mycology/2

25. The yeast form of which dimorphic mold shows a large parent yeast cell surrounded by smaller budding yeast cells?

- A. *Paracoccidioides brasiliensis*
- B. *Sporothrix schenckii*
- C. *Coccidioides immitis*
- D. *Histoplasma capsulatum*

Microbiology/Apply knowledge of fundamental biological characteristics/Mycology/2

Answers to Questions 20–25

- 20. A** *C. tropicalis* and *C. kefyr* (*pseudotropicalis*) differ in their arrangement of blastoconidia along the pseudohyphae. *C. kefyr* (*pseudotropicalis*) forms elongated blastoconidia arranged in parallel clusters that simulate logs in a stream. *Trichosporon* spp. and *Geotrichum* spp. form arthroconidia.
- 21. B** Dimorphic molds are in the yeast form in infected tissues because they are in the yeast form at 37°C. Specimens are cultured and incubated at both room temperature and 35°C to 37°C. To prove that a mold growing at room temperature (or 30°C) is a dimorphic fungus, conversion to the yeast form must be demonstrated via subculture and incubation at 37°C.
- 22. A** The mold form of *C. immitis* shows barrel-shaped arthroconidia separated by empty cells (ghost cells) that cause an uneven staining effect when they are examined under a microscope. *S. schenckii*, *H. capsulatum*, and *B. dermatitidis* produce conidia that are

round or oval in shape in the mold phase.

23. **B** *S. schenckii* is usually acquired by humans through thorns or splinters because it is commonly found on living or dead vegetation. It is called “rose gardener’s disease” because gardeners, florists, and farmers are most often infected. *S. schenckii* is often recovered from exudates of unopened subcutaneous nodules or open draining lesions.
24. **A** *Sepedonium* spp. are saprophytic molds that do not have a yeast phase and produce large spherical tuberculate macroconidia, similar to *H. capsulatum*. Histoplasmosis is a chronic granulomatous infection primarily found in the lungs that invades the reticuloendothelial system. Infection occurs via spores released from decaying bird or chicken droppings that are inhaled when disturbed.
25. **A** *P. brasiliensis* yeast forms are sometimes seen as a “mariner’s wheel” because multiple budding cells completely surround the periphery of the parent cell.

26. **Which group of molds can be ruled out when septate hyphae are observed in a culture?**

- A. Dematiaceus
- B. Mucorales (formerly Zygomycetes)
- C. Dermatophytes
- D. Dimorphic molds

Microbiology/Apply knowledge of fundamental biological characteristics/Mycology/1

27. **Tinea versicolor is a skin infection caused by:**

- A. *Malassezia furfur*
- B. *Trichophyton rubrum*
- C. *Trichophyton schoenleinii*
- D. *Microsporum gypseum*

Microbiology/Apply knowledge of fundamental biological characteristics/Mycology/1

28. **Which of the following structures is (are) invaded by the genus *Trichophyton*?**

- A. Hair
- B. Nails
- C. Skin
- D. All of these options

Microbiology/Apply knowledge of fundamental biological characteristics/Mycology/1

29. **An organism cultured from the skin produces colonies displaying a cherry-red color on Sabouraud dextrose agar after 3 to 4 weeks and teardrop-shaped microconidia along the sides of the hyphae. The most likely identification is:**

- A. *Trichophyton rubrum*
- B. *Trichophyton tonsurans*
- C. *Trichophyton schoenleinii*
- D. *Trichophyton violaceum*

Microbiology/Apply knowledge of fundamental biological characteristics/Mycology/2

30. **Which *Microsporum* species causes an epidemic form of tinea capitis in children?**

- A. *Microsporum canis*
- B. *Microsporum audouinii*
- C. *Microsporum gypseum*

D. All of these options

Microbiology/Correlate clinical and laboratory data/Mycology/2

Answers to Questions 26–30

26. **B** Mucorales (formerly Zygomycetes) commonly recovered from clinical specimens are *Rhizopus* spp. and *Mucor* spp. Both display aseptate hyphae, whereas the other groups above display septate hyphae. Mucorales (Zygomycetes) usually not encountered in clinical specimens are also aseptate and include *Lichtheimia* spp. (formerly *Absidia* spp.), *Rhizomucor* spp., *Cunninghamella* spp., and *Syncephalastrum* spp.
27. **A** *M. furfur* has a worldwide distribution and causes a superficial, brownish, dry, scaly patch on the skin of light-skinned persons and lighter patches on persons with dark skin. *M. furfur* is not cultured because diagnosis can be made from microscopic examination of the skin scales. Skin scrapings prepared in KOH show oval- or bottle-shaped cells that exhibit monopolar budding in the presence of a cell wall and also produce small hyphae.
28. **D** *Trichophyton* spp., *Microsporum* spp., and *Epidermophyton* spp. are the organisms causing human dermatomycoses or cutaneous infections. *Trichophyton* spp. infect hair and nails, as well as skin. Infections with members of the genus *Microsporum* are confined to the hair and skin, whereas infections caused by the genus *Epidermophyton* are seen only on skin and nails.
29. **A** Members of the genus *Microsporum* produce club-shaped microconidia and are usually pigmented white, buff, yellow, or brown. *Epidermophyton* does not display microconidia and produces yellow-green or yellow-tan colonies. *T. rubrum* can be differentiated from the other members of the genus by its distinctive cherry-red color. *Trichophyton mentagrophytes* may also produce a red pigment, but it is usually rose colored or orange, or deep red. *T. tonsurans* produces white-tan to yellow, suede-like colonies. *T. schoenleinii* produces white to cream-colored colonies, and *T. violaceum* produces port wine to deep violet colonies.
30. **B** *M. audouinii* and *T. tonsurans* may both cause epidemic tinea capitis in children. *M. audouinii* causes a chronic infection transmitted directly via infected hairs on caps, hats, combs, upholstery, and hair clippers. Infected hair shafts fluoresce yellow-green under a Wood lamp. *M. audouinii* does not usually sporulate in culture and forms atypical vegetative forms, such as antler and racquet hyphae and terminal chlamydoconidia. In contrast, *M. canis* produces spindle-shaped, thick-walled multicelled macroconidia, and *M. gypseum* produces ellipsoidal, multicellular macroconidia.
31. **Microscopic examination of a fungus cultured from a patient with athlete's foot showed large, smooth-walled, club-shaped macroconidia appearing singly or in clusters of two to three from the tips of short conidiophores. The colonies did not produce microconidia. What is the most likely identification?**
- A. *Trichophyton* spp.
B. *Alternaria* spp.
C. *Epidermophyton* spp.

D. *Microsporum* spp.

Microbiology/Evaluate laboratory data to make identifications/Mycology/2

32. Which *Trichophyton* species causes the favus type of tinea capitis seen in the Scandinavian countries and in the Appalachian region of the United States?

A. *Trichophyton verrucosum*

B. *Trichophyton violaceum*

C. *Trichophyton tonsurans*

D. *Trichophyton schoenleinii*

Microbiology/Correlate clinical and laboratory data/Mycology/2

33. The hair baiting test is used to differentiate which two species of *Trichophyton* that produce red colonies on Sabouraud agar plates?

A. *Trichophyton mentagrophytes* and *Trichophyton rubrum*

B. *Trichophyton tonsurans* and *Trichophyton schoenleinii*

C. *Trichophyton tonsurans* and *Trichophyton violaceum*

D. *Trichophyton verrucosum* and *Trichophyton rubrum*

Microbiology/Correlate clinical and laboratory data/Mycology/2

34. A mold that produces colonies with a dark brown, green-black, or black appearance of both the surface and reverse side is classified as a:

A. Dematiaceous mold

B. Dermatophyte

C. Hyaline mold

D. Dimorphic fungus

Microbiology/Apply knowledge of fundamental biological characteristics/Mycology/1

35. A rapidly growing hyaline mold began as a white colony but soon developed a black “pepper” effect on the agar surface. The older colony produced a black matte, making it resemble a dematiaceous mold. What is the most likely identification?

A. *Penicillium notatum*

B. *Aspergillus niger*

C. *Paecilomyces* spp.

D. *Scopulariopsis* spp.

Microbiology/Apply knowledge of fundamental biological characteristics/Mycology/2

36. Which dematiaceous mold forms flask-shaped phialides, each with a flask-shaped collarette?

A. *Phialophora* spp.

B. *Exophiala* spp.

C. *Wangiella* spp.

D. All of these options

Microbiology/Apply knowledge of fundamental biological characteristics/Mycology/2

Answers to Questions 31–36

- 31. C** *Epidermophyton* spp. do not produce microconidia; this differentiates them from *Trichophyton* spp. and *Microsporum* spp. *Alternaria* is not a dermatophyte. *Epidermophyton floccosum* is the most frequently isolated member of the genus and infects skin, but not hair.

32. **D** *T. schoenleinii* is identified microscopically by its characteristic antler-shaped hyphae and chlamydoconidia in the absence of conidia.
33. **A** *T. mentagrophytes* may produce a deep red pigment seen through the reverse side of the agar plate that resembles the cherry-red pigment produced by *T. rubrum*. However, *T. mentagrophytes* can be differentiated by its ability to invade the hair shaft. *T. rubrum* grows on the surface of hair but does not penetrate the shaft.
34. **A** The dematiaceous molds are easily recognized and confirmed by observing dark-yellow or brown septate hyphae upon microscopic examination.
35. **B** *A. niger* is the only species listed producing black conidia, which causes a “pepper” effect as the colony grows. The reverse side of the agar plate remains buff or cream colored, which differentiates it from the dematiaceous (dark) molds.
36. **A** *Phialophora*, *Exophiala*, and *Wangiella* all produce phialides, but the last two genera form elongated, tubelike phialides without a collarette, as opposed to the flask-shaped phialides of *Phialophora*, which contain clusters of conidia at the tips.

37. Which *Aspergillus* species, recovered from sputum or bronchial mucus, is the most common cause of pulmonary aspergillosis?

- A. *Aspergillus niger*
- B. *Aspergillus flavus*
- C. *Aspergillus fumigatus*
- D. All of these options

Microbiology/Correlate clinical and laboratory data/Mycology/2

38. A hyaline mold recovered from a patient with AIDS produced rose-colored colonies with lavender centers on Sabouraud dextrose agar. Microscopic examination showed multiseptate macroconidia appearing as sickles or canoes. What is the most likely identification?

- A. *Fusarium* spp.
- B. *Wangiella* spp.
- C. *Exophiala* spp.
- D. *Phialophora* spp.

Microbiology/Evaluate laboratory data to make identifications/Mycology/3

39. Material from a fungus-ball infection produced colonies with a green surface on Sabouraud agar in 5 days at 30°C. Microscopic examination showed club-shaped vesicles with sporulation only from the top half of the vesicle. This hyaline mold is most probably which *Aspergillus* spp.?

- A. *Aspergillus niger*
- B. *Aspergillus fumigatus*
- C. *Aspergillus flavus*
- D. *Aspergillus terreus*

Microbiology/Evaluate laboratory data to make identifications/Mycology/3

40. A rapidly growing nonseptate mold produced colonies with a gray surface resembling cotton candy that covered the entire plate. Microscopic examination revealed sporangiophores arising between, not opposite, the rhizoids and producing pear-shaped sporangia. What is the most likely identification?

- A. *Lichtheimis* (formerly *Absidia*) spp.
- B. *Penicillium* spp.
- C. *Rhizopus* spp.
- D. *Aspergillus* spp.

Microbiology/Evaluate laboratory data to make identifications/Mycology/3

- 41. An India ink test was performed on CSF from a male patient with HIV infection. Many encapsulated yeast cells were seen in the centrifuged sample. Further testing revealed a positive urease test and growth of brown colonies on niger seed agar. The diagnosis of meningitis was caused by which yeast?**

- A. *Candida albicans*
- B. *Cryptococcus neoformans*
- C. *Cryptococcus laurentii*
- D. *Candida tropicalis*

Microbiology/Apply knowledge for identification/Mycology/3

Answers to Questions 37–41

- 37. C** *A. fumigatus* is most often associated with compost piles and is found in the soil of potted plants. *A. niger* is the cause of cavitory fungus ball lesions of the lungs and nasal passages.
- 38. A** *Fusarium* spp. are usually a contaminant but are sometimes seen as a cause of mycotic eye, nail, or skin infection in debilitated patients. *Fusarium* is a hyaline (light) mold and grows on Sabouraud agar plates at 30°C within 4 days. The other three organisms are members of the Dematiaceae family (dark molds).
- 39. B** *A. fumigatus* is the most common cause of aspergillosis. It is characterized by sporulation only from the upper half or two-thirds of the vesicle. Colonies of *A. niger* are white with black pepper growth and produce phialides over the entire vesicle, forming the classic “radiate” head. *A. flavus* colonies are yellow to yellow-green and produce phialides that cover the entire vesicle and point out in all directions. *A. terreus* produces brown colonies and phialides that also cover the entire vesicle.
- 40. A** *Lichtheimia* (*Absidia*) spp. are similar to *Rhizopus* spp. except for the location of rhizoids (rootlike hyphae). The rhizoids of *Rhizopus* spp. are located at the point where the stolons and sporangiophores meet, whereas those of *Lichtheimia* (*Absidia*) spp. arise at a point on the stolon between the rhizoids. *Penicillium* spp. and *Aspergillus* spp. do not form rhizoids.
- 41. B** Immunocompromised patients are at risk for invasion of *C. neoformans*. The polysaccharide capsule of *C. neoformans* is not recognized by phagocytes, which cause patients with impaired cell-mediated immunity to become readily infected with *C. neoformans*.

42. A bone marrow sample obtained from an immunocompromised patient revealed small intracellular cells using a Wright stain preparation. Growth on Sabouraud–dextrose agar plates of a mold phase at 25°C and a yeast phase at 37°C designates the organism as dimorphic. The mold phase produced thick, spherical tuberculated

macroconidia. What is the most likely identification?

- A. *Histoplasma capsulatum*
- B. *Sepedonium* spp.
- C. *Sporothrix schenckii*
- D. *Coccidioides immitis*

Microbiology/Apply knowledge for identification/Mycology/3

- 43. A lung biopsy obtained from an immunocompromised patient showed many “cup-shaped” cysts (gray to black) in a foamy exudate (green background) using Gomori methenamine silver (GMS) stain. The organism cannot be cultured because it does not grow on routine culture media for molds. The patient was diagnosed with pneumonia resistant to antibiotic treatment. The most likely identification is:**

- A. *Pneumocystis jirovecii* (previously *Pneumocystis carinii*)
- B. *Histoplasma capsulatum*
- C. *Sporothrix schenckii*
- D. *Scopulariopsis* spp.

Microbiology/Apply knowledge for identification/Mycology/3

- 44. Upon direct examination of a sputum specimen, several spherules that contained endospores were noted. Growth on Sabouraud–dextrose agar showed aerial mycelial elements. The septate hyphae produced barrel-shaped arthroconidia. What is the most likely identification?**

- A. *Talaromyces* (formerly *Penicillium*) *marneffei*
- B. *Scopulariopsis* spp.
- C. *Cryptococcus neoformans*
- D. *Coccidioides immitis*

Microbiology/Evaluate laboratory data to make identifications/Mycology/3

Answers to Questions 42–44

- 42. A** Thermally dimorphic *H. capsulatum* produce microconidia and hyphal fragments at 37°C (yeast phase), whereas at 25°C (mold phase), the organism displays large, thick-walled, round macroconidia with knobby or knoblike projections. The yeast form is able to survive within circulating monocytes or tissue macrophages that can be demonstrated with Giemsa or Wright stain.
- 43. A** *P. jirovecii* (*P. carinii*) is classified as a fungus but formerly was classified as a parasite. It is best recovered by bronchoalveolar lavage or induced sputum in immunocompromised patients. Open lung biopsy sample was the specimen of choice before the AIDS epidemic. GMS stain is used to identify the organism which stains the cyst form in lung tissue, but not the trophic form. By using hematoxylin and eosin (H&E) staining on a smear, an eosinophilic, foamy material can be detected which is formed from the successful replication of the organism in alveolar lining fluid.
- 44. D** *C. immitis* endospores are often confused with yeast cells, but they do not bud. *C. immitis* is endemic in the southwestern United States. Because arthroconidia are *highly infectious*, an open plate should *not* be used, and a slide culture test should *not* be performed. Rather, tubed media is used for testing, and all work should be performed in a biological safety cabinet.

45. A bone marrow specimen was obtained from an immunocompromised patient who tested positive for HIV. The organism grew rapidly at 3 days showing a mold form (at 25°C), displaying conidiophores with four to five terminal metulae with each having four to six phialides. The conidia at the end of the phialides were oval and in short chains. They appear as a fan or broom when viewing under 10× and 40× magnification. At 37°C, the yeast form grew more slowly, showing conidia that formed hyphal elements breaking at the septa to produce oval arthroconidia. This thermodimorphic mold is *most* likely:

- A. *Paecilomyces* spp.
- B. *Talaromyces* (formerly *Penicillium*) *marneffe*
- C. *Rhizomucor* spp.
- D. *Aspergillus fumigatus*

Microbiology/Evaluate laboratory data to make identifications/Mycology/3

46. What is the specimen of choice for the initial diagnosis of *P. jirovecii* (*carinii*) in an immunocompromised patient, such as someone with AIDS?

- A. Induced sputum
- B. Open-thorax lung biopsy
- C. CSF
- D. Urine

Microbiology/Specimen collection/Mycology/2

47. A transplant patient is suspected of having invasive aspergillosis on the basis of clinical and radiological findings. Which specimen is *best* for the initial identification of aspergillosis by soluble antigen testing?

- A. Blood culture
- B. Lung biopsy
- C. Serum or urine
- D. Sputum

Microbiology/Specimen collection/Mycology/2

48. What is the *most* common cause of mucormycosis infection in humans?

- A. *Penicillium* spp.
- B. *Candida albicans*
- C. *Scopulariopsis* spp.
- D. *Rhizopus* spp.

Microbiology/Identification/Mycology/2

Answers to Questions 45 - 48

45. **B** Other *Penicillium* spp. are differentiated from *Talaromyces* (*Penicillium*) *marneffe* (thermally dimorphic) through conversion from the mold to yeast phase. *Talaromyces* (*P.*) *marneffe* are seen as yeast at 35°C to 37°C on 5% sheep blood agar or in BHI broth. Other *Penicillium* spp. do not display a yeast phase. *T. marneffe* are recovered from the blood, skin, lymph nodes, bone marrow, and internal organs of immunocompromised patients.

46. **A** Open lung biopsy is *not* recommended for persons with AIDS because of the increased risk of death, infection, and complications associated with this surgical

procedure. Persons with AIDS and *P. jirovecii* (*carinii*) pneumonia (PCP) can be diagnosed with chest radiography, special stains, and PCR methods from specimens obtained by bronchoalveolar lavage (BAL). Patients without AIDS have a much lower yield, and therefore, besides BAL, tracheal aspiration, endoscopic lung biopsy, cellular material from bronchial brushings, and open lung biopsy specimens are more often indicated.

47. **C** To determine an early diagnosis, soluble antigens (the galactomannan assay targets antigens of *Aspergillus* spp.) are detected by using such methods as ELISA, latex agglutination, and immunoblotting using serum or body fluids. Blood and sputum cultures are usually negative, and tissue biopsy is associated with a significantly higher risk of complications. Nucleic acid amplification assays, multiplex amplification and reverse transcription PCR (RT-PCR) have been developed which detect systemic aspergillosis in respiratory specimens. MALDI-TOF MS methodology has proven successful on old and young cultures, with 98.6% correct identification.
48. **D** Vascular invasion, followed by thrombosis, tissue infarction, and necrosis are the most common clinical manifestations of cutaneous infections caused by *Rhizopus* spp. Infections occur mainly in patients who are already being treated for other conditions, including lymphoma, leukemia, neutropenia, renal failure, and diabetic ketoacidosis.

49. A thermally dimorphic fungus shows a filamentous mold form with tuberculate macroconidia at room temperature, and a yeast form above 35°C. Which organism best fits this description?

- A. *Histoplasma capsulatum*
- B. *Paracoccidioides brasiliensis*
- C. *Candida albicans*
- D. *Coccidioides immitis*

Microbiology/Identification/Mycology/2

50. SITUATION: After a vacation to the southwestern United States, a midwesterner complained of flulike symptoms with fever, chills, nonproductive cough, and chest pain. Microscopic examination of sputum, cleared with KOH, revealed large, thick-walled spherules containing endospores. Upon culture, the mold phase showed septate hyphae and alternating barrel-shaped arthroconidia. Which organism is most likely the cause of this pneumonia?

- A. *Coccidioides immitis*
- B. *Histoplasma capsulatum*
- C. *Paracoccidioides brasiliensis*
- D. *Talaromyces* (formerly *Penicillium*) *marneffei*

Microbiology/Identification/Mycology/3

Answers to Questions 49–50

49. **A** The yeast phase of *H. capsulatum* develops as small oval budding cells seen in macrophages. The yeast phase of *P. brasiliensis* develops large cells that bud creating a structure resembling a mariner's wheel, and *C. immitis* produces very large spherules. At 25°C, *P. brasiliensis* produces small single conidia, and *C. immitis*

produces arthroconidia. *Candida* spp. are not dimorphic. *H. capsulatum* is found in soil containing excrement of birds and bats. It is endemic in the Ohio and Mississippi River valleys, and the most common systemic mycosis in North America.

50. **A** *C. immitis* is a soil fungus and is endemic to the southwestern United States. Infection results from inhaling the arthroconidia, which form endospore-forming spherules and leads to infection within 1 to 3 weeks. Nonimmunocompromised people usually do not require treatment and are immune to reinfection. However, immunodeficient persons may develop symptomatic pulmonary and multiorgan infections.

51. **Skin scrapings from a young male patient showed many yeast cells of oval and bottle-shapes with monopolar budding. The direct microscopic examination also showed hyphal elements with a “spaghetti and meatball” morphological appearance. The most likely identification of this yeast is?**

A. *Malassezia furfur*
B. *Candida glabrata*
C. *Candida albicans*
D. *Cryptococcus neoformans*

Microbiology/Identification/Mycology/2

52. **Microscopically, *Scopulariopsis* spp. resemble *Penicillium* spp. Which of the following best differentiates the two?**

A. Nonseptate hyphae and foot cells
B. Brush-like conidiophores and large echinulate conidia
C. No hyphae and no phialides
D. Growth at 45°C.

Microbiology/Characteristics/Identification/Mycology/2

53. **A fast-growing (2–5 days) culture of fluffy white to blue-green colonies from a respiratory specimen obtained from an immunocompromised ICU patient showed septate hyphae with many conidiophores. The tips of the conidiophores expand into large dome-shaped vesicles with bottle-shaped phialides covering the upper half or two-thirds of its surface. Long chains of small conidia form a columnar mass on the surface. At the base of the conidiophore, a “foot cell” is seen attaching the conidiophore to the septate hyphae. The most likely organism is?**

A. *Penicillium* spp.
B. *Scopulariopsis* spp.
C. *Aspergillus fumigatus*
D. *Paecilomyces* spp.

Microbiology/Culture characteristics for Identification/Mycology/3

Answers to Questions 51–53

51. **A** *M. furfur*, the cause of tinea versicolor, is often diagnosed from a direct skin scraping specimen. The classic “spaghetti and meatball” formations of the yeast and hyphal cells are distinct. In fungemia, the blood culture specimen will show small yeast cells without pseudohyphae.
52. **B** Microscopically, *Scopulariopsis* spp. resemble *Penicillium* spp. except for the

following: *Scopulariopsis* spp. produce annellophores with flask-shaped annelides bearing large lemon-shaped, rough-walled conidia in chains; *Penicillium* spp. display brush-like conidiophores producing metulae with flask-shaped phialides, resulting in chains of small, smooth-walled conidia.

53. C A “foot cell” is characteristic of *Aspergillus fumigatus*. The presence of a foot cell, which is a T- or L-shaped cell at the base of the conidiophore (it is not a separate cell), is characteristic and distinct for *A. fumigatus* which is thermotolerant, growing at temperatures up to 45°C.

8.10

Virology

1. Classification of viruses is made by:

- A. Complement fixation serology
- B. Electron microscopy
- C. Nucleic acid composition
- D. Cellular inclusion bodies

Microbiology/Apply knowledge of fundamental biological characteristics/Viruses/1

2. Which virus is the *most common* etiological agent of viral respiratory diseases in infants and children?

- A. Respiratory syncytial virus (RSV)
- B. Measles virus
- C. Coxsackie A virus
- D. Coxsackie B virus

Microbiology/Apply knowledge of fundamental biological characteristics/Viruses/1

3. The *most common* viral syndrome of pericarditis, myocarditis, and pleurodynia (pain upon breathing) is caused by:

- A. Herpes simplex virus (HSV)
- B. Respiratory syncytial virus
- C. Epstein-Barr virus (EBV)
- D. Coxsackie B virus

Microbiology/Apply knowledge of fundamental biological characteristics/Viruses/1

4. Which of the following viruses is implicated along with EBV as a cause of infectious mononucleosis?

- A. Cytomegalovirus (CMV)
- B. Coxsackie A virus
- C. Coxsackie B virus
- D. Hepatitis B virus

Microbiology/Apply knowledge of fundamental biological characteristics/Viruses/1

5. The *most common* causes of viral pneumonia in adults are:

- A. Influenza and adenovirus
- B. Hepatitis A and B viruses
- C. Coxsackie A and B viruses
- D. Herpes simplex and CMV

Microbiology/Apply knowledge of fundamental biological characteristics/Viruses/1

Answers to Questions 1–5

- 1. C True viruses have nucleic acid that is either RNA or DNA, and this serves as the basis for initial classification. Members of these classes are further divided into groups that

cause human disease on the basis of mode of transmission, tissues invaded, diseases produced, and antigenic characteristics.

2. **A** RSV is the cause of croup, bronchitis, bronchiolitis, and interstitial pneumonia. Hospitalized children age less than 1 year are the most susceptible group.
3. **D** Coxsackie A virus, Coxsackie B virus, and the echoviruses are most commonly implicated in myocarditis and other syndromes, including acute cerebellar ataxia and hepatitis. Like poliovirus, infections are more common in the summer and fall seasons and gain entry through the gastrointestinal tract.
4. **A** CMV infection in a previously healthy individual causes a self-limiting mononucleosis syndrome. CMV is an opportunistic pathogen that may produce lifelong infections and causes a variety of diseases, including congenital and neonatal infection, hepatitis, pneumonia, and disseminated infection in immunocompromised patients.
5. **A** Influenza and adenoviruses are the main causes of respiratory infections, including the common cold, tracheobronchitis, and pneumonia. Adenoviruses also cause conjunctivitis, keratitis, cystitis, and gastroenteritis and are transmitted by aerosolized droplet, contaminated respiratory secretions, stool or fomites.

6. Which virus belonging to the Reoviridae group causes gastroenteritis in infants and young children but an asymptomatic infection in adults?

- A. Coxsackie B virus
- B. Rotavirus
- C. Respiratory syncytial virus
- D. Rhabdovirus

Microbiology/Apply knowledge of fundamental biological characteristics/Viruses/1

7. A very small, single-stranded DNA virus that causes a febrile illness with a rash, and is called the *fifth childhood disease* after rubeola, rubella, varicella, and roseola is:

- A. Rotavirus
- B. Adenovirus type 40
- C. Coxsackie A virus
- D. Parvovirus B19

Microbiology/Apply knowledge of fundamental biological characteristics/Viruses/1

8. Hepatitis B virus can be transmitted by:

- A. Acupuncture
- B. Tattoos
- C. Sexual contact
- D. All of these options

Microbiology/Apply knowledge of fundamental biological characteristics/Viruses/1

9. Which virus has been implicated in adult gastroenteritis resulting from ingestion of contaminated food (especially shellfish) and water?

- A. Noroviruses (Norwalk-like viruses)
- B. Rotavirus
- C. Hepatitis C virus
- D. Coronavirus

Microbiology/Apply knowledge of fundamental biological characteristics/Viruses/1

10. Which virus is associated with venereal and respiratory tract warts and produces lesions of skin and mucous membranes?

- A. Polyomavirus
- B. Poxvirus
- C. Adenovirus
- D. Papillomavirus

Microbiology/Apply knowledge of fundamental biological characteristics/Viruses/1

11. A clinical test used for the detection and identification of viral infections other than culture is:

- A. Histology/cytology
- B. Polymerase chain reaction/DNA probes
- C. Serology
- D. All of these options

Microbiology/Apply principles of basic laboratory procedures/Viruses/1

Answers to Questions 6–11

- 6. B** Rotaviruses have been implicated in both nosocomial infections and epidemic gastroenteritis. Children ages 3 to 24 months are most commonly affected. Diarrhea begins after an incubation period of 3 days, lasts for 2 to 10 days and is associated with vomiting and dehydration. In immunosuppressed children, rotavirus causes a chronic infection.
- 7. D** Parvovirus causes a fever and characteristic “slapped cheek” rash in young children. Adults are usually immune, but immunocompromised persons may exhibit arthritis or anemia (the virus infects immature RBCs in bone marrow).
- 8. D** Although the most common mode of transmission of hepatitis B is via needle puncture, it may also be transmitted by other parenteral means, including through sexual transmission and contact with contaminated blood via broken skin or mucous membranes.
- 9. A** Noroviruses are small RNA viruses that have been implicated in epidemics of community gastroenteritis as well as sporadic infections. Unlike rotaviruses, which cause gastroenteritis in infants and young children, noroviruses produce infections in all age groups.
- 10. D** Human papillomaviruses (HPVs) cause genital warts, the most prevalent sexually transmitted viral disease in the United States. Several strains, including HPV-6, HPV-11, HPV-16, and HPV-18, are associated with penile, cervical, and vaginal neoplasia, causing more than 60% of the cancers. Because the virus cannot be cultured in vitro, diagnosis is usually made by using DNA probes, histopathology, or cytological examination of cutaneous biopsy of cells. A diagnostic characteristic of infected cells is koilocytosis, a perinuclear clearing in the squamous epithelium accompanied by nuclear atypia. Currently, the FDA has licensed three vaccines for HPV: Cervarix, Gardasil, and Gardasil-9.
- 11. D** In addition to serological tests for antibodies against the virus and DNA probes that identify viral DNA or RNA, cytological and histological examination aid in the rapid

diagnosis of several viruses. The rubella virus is detected serologically by testing for specific IgG and IgM antibodies. Testing is routinely done on pregnant women, as a prenatal precaution.

12. Which technique is used for the confirmation of infection with HIV-1 and HIV-2?

- A. Western blot (immunoblot) assay
- B. ELISA
- C. FDA-approved NAT
- D. p24 Antigen testing

Microbiology/Select methods/Reagents/Media/Viruses/2

13. A 13-year-old boy was admitted to the hospital with a diagnosis of viral encephalitis. History revealed that the boy harbored wild raccoons from nearby woods. What is the best method to determine if the boy has contracted rabies?

- A. Remove the brainstems from all of the raccoons and examine for cytopathic effects
- B. Request immunofluorescent test for antibody on the saliva from all of the raccoons
- C. Request immunofluorescent test for antigen in cutaneous nerve specimen obtained by nuchal biopsy
- D. Isolate the virus from the saliva of both the animals and the patient

Microbiology/Select methods/Reagents/Media/Viruses/3

14. A 65-year-old woman was admitted to the hospital with acute respiratory distress, fever, myalgia, and headache. Influenza A or B was suspected after ruling out bacterial pneumonia. Which of the following methods could be used to confirm influenza infection?

- A. EIA
- B. RT-PCR
- C. Direct examination of nasal epithelium for virus using fluorescent antibody stain
- D. All of these options

Microbiology/Select methods/Reagents/Media/Viruses/3

Answers to Questions 12–14

- 12. C** The Western blot assay is no longer used to confirm a positive serological test of antibodies to HIV. Western blot detects antibodies but it may miss people in the window phase of infection, is not confirmatory for neonatal infections because of the presence of maternal antibodies, and has a long turnaround time resulting in loss of contact with the patient. For these reasons, the CDC recommends a new approach using a simultaneous test for antibodies to both HIV-1 and HIV-2 and p24 antigen to screen for infection by immunoassay. Positive specimens would be tested by separate immunoassays to identify whether the infection is HIV-1 or HIV-2. Samples testing negative in suspected cases should be tested by a nucleic acid amplification method, such as PCR for viral RNA for definite identification.
- 13. C** Using direct immunofluorescence microscopy (DIF), rabies antigen can be detected in the cutaneous nerves surrounding the hair follicles of the posterior region of the neck (nuchal biopsy) and in the epithelial cells obtained through corneal impression. Antibodies to rabies can be detected in the serum and CSF of infected persons within 8 to 10 days of illness; however, infection usually occurs several months before the onset

of symptoms. Isolation of virus from the saliva of the patient may be accomplished by mouse inoculation or by inoculation of susceptible cell culture lines with subsequent detection by immunofluorescent antibodies. Postmortem diagnosis of rabies is accomplished by using direct immunofluorescent assay of brain tissue. Brain tissue testing from the dead animal results in prompt treatment/prophylaxis of the human victim after an animal bite or injury.

- 14. D** Influenza virus types A, B, and C may be grown and isolated in cell cultures by using primary monkey kidney (PMK) cells permitting identification within 1 to 3 days in specialized laboratories performing viral cultures. Direct fluorescent and enzyme immunoassays by using monoclonal antibodies to nucleoprotein antigens in infected nasal epithelium are used for rapid diagnosis of both influenza A and influenza B infections. Influenza C testing is not routinely requested, and this type is referred to as “mild” influenza. RT-PCR is replacing the cell culture method because of better turn-around time, from collection of specimen to identification. It is also becoming the gold-standard for identifying respiratory viruses.

- 15. The most rapid definitive diagnosis of a genital herpes simplex (HSV-2) infection in a 20-year-old man is made by using which method?**

A. Direct immunofluorescence test for viral antigen in vesicle fluid
B. Titer of serum and seminal fluid for antibodies to herpes simplex
C. Detection of anti-herpes simplex in seminal fluid
D. Cell culture of vesicle fluid

Microbiology/Select methods/Reagents/Media/Viruses/2

- 16. A 20-year-old female college student complained of a sore throat and extreme fatigue. The physician noted lymphadenopathy and ordered a rapid test for infectious mononucleosis antibodies that was negative. Bacterial cultures were negative, as were serological tests for influenza A and B, HIV-1, CMV, hepatitis B, and antistreptolysin O. What would be the next line of viral testing to establish a diagnosis?**

A. Herpes simplex
B. Rubella
C. Epstein-Barr
D. West Nile

Microbiology/Select testing for identification/Virology/3

- 17. A 60-year-old male gardener from New York State was hospitalized with flulike symptoms and was eventually diagnosed with encephalitis. While working in his garden, he had noticed several dead birds around his bird feeder. The region is known for heavy mosquito infestation. What is the most likely cause of his illness?**

A. West Nile virus
B. Epstein-Barr virus
C. Parvovirus
D. Hantavirus

Microbiology/Select diagnosis/Virology/3

- 18. A 30-year-old male patient, who was a contractor and building inspector in the southwestern United States, complained of difficulty breathing and was admitted to the hospital with severe respiratory disease. The physician noted a high fever and cough.**

Two days earlier, the patient had inspected an old abandoned warehouse that was infested by rodents. The patient was given intravenous antibiotics, but 2 days into therapy, the pneumonia worsened, leading to pulmonary edema. Which organism should be suspected of causing his illness?

- A. Hantavirus
- B. Rotavirus
- C. West Nile virus
- D. Norovirus (formerly Norwalk-like virus)

Microbiology/Select diagnosis/Virology)3

Answers to Questions 15–18

- 15. A** DIF testing of vesicle (lesion) fluid for virus using fluorescein-conjugated antibodies is the most rapid method for diagnosis of genital herpes infection. Immunofluorescence and immunoperoxidase methods are also used to distinguish HSV-1 and HSV-2. PCR and real-time PCR are more sensitive rapid methods for detection and identification. Viral cell culture is also very sensitive and may yield a positive result within 24 hours when fluid contains a high concentration of virus. Vero cells or primary human embryonic cells are inoculated with vesicle fluid and examined for cytopathic effects (CPEs), the most common of which are large “balloon” cells and multinucleated giant cells.
- 16. C** EBV serological testing for IgM-VCA (viral capsid antigen) during the illness may or may not be positive. Patients who present with an infectious mononucleosis–like syndrome should be tested for both EBV and CMV. Both viruses cause the same symptoms during the acute phase of the illness.
- 17. A** West Nile virus causes neurological diseases with meningitis and encephalitis at the top of the list. The animal reservoirs are birds, with humans being accidental hosts. Transmission of West Nile virus is from mosquito to bird.
- 18. A** Hantavirus is transmitted by a rodent host, the deer mouse, was originally endemic in the southwest United States, but currently has been discovered throughout the country. The name of the hantavirus responsible for outbreaks in the southwest is the Sin Nombre virus. Breathing in excrement of mice is the most common mode of infection, and the lung is the site of initial infection, causing labored breathing or hantavirus pulmonary syndrome (HPS). Diagnosis is usually made by using an IgM/IgG ELISA assay and RT-PCR. Isolation of the virus by culture has not been accomplished to date. The primary site of infection for norovirus and rotavirus is the gastrointestinal area and for hantavirus the pulmonary sector.
- 19. A** 3-year-old female child was admitted to the hospital following a 2-day visit with relatives over the Christmas holidays. Vomiting and diarrhea left her severely dehydrated. No other members of the family were affected. All bacterial cultures proved negative. A stool sample should be tested for which virus?
- A. CMV
 - B. EBV
 - C. Hepatitis D

D. Rotavirus

Microbiology/Select testing for identification/Virology/3

20. A 25-year-old male patient was diagnosed with HIV-1 by using enzyme immunoassay; the patient tested positive twice, and the diagnosis was confirmed with RT-PCR. Which laboratory test(s) should be performed prior to initiating antiviral therapy and for monitoring during antiviral therapy?
- A. Quantitative plasma virus concentration (viral load testing)
 - B. Quantitation of CD4 lymphocytes
 - C. Phenotype/genotype resistance testing
 - D. All of these options

Microbiology/Select tests/Virology/3

21. A 6-month-old male infant was hospitalized with a respiratory infection. He was diagnosed with apnea and bronchiolitis. Further testing revealed congenital heart disease. Bacterial cultures were negative for *S. pneumoniae* and *H. influenzae*. What viruses should be tested?
- A. RSV
 - B. Rotavirus
 - C. Norovirus (Norwalk-like virus)
 - D. HIV

Microbiology/Select testing for identification/Viruses/3

22. A young male hunter encountered a fox in his path during a walk in the woods. The fox was staggering but appeared nonthreatening. The man tried to avoid contact but was attacked and bitten on the leg. The bite broke the skin but was not deep. Wildlife officials were unable to locate the fox for testing. What procedure should be performed next for the hunter?
- A. Spinal tap with CSF testing for rabies virus (within 2 months)
 - B. Administration of hyperimmune antirabies globulin and rabies vaccine
 - C. Biopsy of the wound site
 - D. Throat culture and blood culture (weekly)

Microbiology/Evaluate information for testing and identification/Virology/3

Answers to Questions 19–22

19. **D** Rotavirus is one of the most common causes of gastroenteritis in infants and young children (ages 6 months to 2 years). Vomiting and diarrhea are also common symptoms of norovirus infections, but the prevalence of rotavirus during the winter months and the absence of illness in other family members point to rotavirus being a more likely cause. Commercial availability of immunoassays for rotavirus makes its diagnosis easier to establish and rule out than infection with norovirus.
20. **D** The decision to initiate antiviral therapy is based on the presence or absence of symptoms, CD4 lymphocyte count, and the viral load. For example, treatment is usually withheld from patients with CD4 counts greater than 350/ μ L and viral load less than 55,000/mL and is instituted in asymptomatic patients if the CD4 count is less than 200/ μ L, regardless of viral load. Treatment failure within the first year with three-drug regimens is 35% to 45%, and drug resistance testing (genotype and/or phenotype

testing) is recommended to identify drug-resistant strains prior to initiating treatment. For successful antiviral therapy, the HIV plasma viral load should be reduced to undetectable levels.

21. **A** RSV is spread by large particles, such as dust, and is one of the most common causes of hospitalization for respiratory illness in infants age less than 1 year. RSV causes bronchiolitis, pneumonia, and croup in infants and upper respiratory illness in children. It has also been found to cause nosocomial infection in nursing homes. Diagnosis is made by using EIA, fluorescent antibody (FA) staining, and cell culture (HEp-2).
22. **B** Rabies virus can be detected by FA staining and PCR testing. The virus replicates at the site of the bite and penetrates the surrounding tissue, finding its way to the central nervous system. Because the source cannot be tested, the best immediate course of action is to initiate postexposure prophylaxis with antirabies globulin and to immunize the patient with the rabies vaccine.

23. **A 40-year-old female presented with a respiratory infection after returning home from a visit to her homeland of China. Rapid onset of pneumonia in the lower respiratory area prompted the physician to place her in isolation. She was diagnosed presumptively with severe acute respiratory syndrome (SARS) and placed on a respirator. What type of testing should be done next to diagnose this disease?**

A. RT-PCR
B. Latex agglutination test
C. Blood culture
D. Complement fixation

Microbiology/Select tests for identification/Virology/3

24. **A 25-year-old pregnant female with genital lesions delivered a premature newborn with complications. The baby tested negative for bacterial infection (cultures of blood and urine). Antigen testing of the baby's urine proved negative for group B streptococci and *S. pneumoniae*. The mother tested negative for bacterial sexually transmitted diseases (STDs) and for group B streptococci. The baby was treated with acyclovir but failed to survive. What was the *most likely* cause of death?**

A. CMV
B. HIV
C. RSV
D. HSV

Microbiology/Select diagnosis/Viruses/3

25. **A young father of two small children complained of a rash on the torso of his body. The children had been diagnosed with chickenpox and confined to their home. The father had experienced chickenpox as a child and knew he did not have the same rash as his children. What is the *most likely* cause of the father's rash?**

A. HSV 1
B. Varicella-zoster virus
C. HSV 2
D. EBV

Microbiology/Select diagnosis/Viruses/3

26. **The organs of a 65-year-old male were donated soon after a fatal auto accident. Three**

recipients of his corneas and kidneys died within weeks after receiving his organs. Which of the following viruses *most likely* caused the death of these recipients?

- A.** Human immunodeficiency virus
- B.** Rabies
- C.** Norovirus (Norwalk-like virus)
- D.** Rotavirus

Microbiology/Select diagnosis/Viruses/3

Answers to Questions 23–26

- 23. A** SARS virus was first identified in China in November 2002 when it caused a worldwide outbreak that was finally contained by July 2003. The virus belongs to the group of coronaviruses causing the common cold and is easily transmitted to health care workers who have close contact with infected individuals. This virus is the cause of a severe lower respiratory infection that can be fatal. Laboratory confirmation may be done by using the recommended method, that is, RT-PCR testing, which is available commercially. Typically, PCR is used on two different specimen types or the same specimen type submitted at least 2 days apart. If both tests are positive, the infection is confirmed. Cell culture (Vero-E6 cell line) can be performed if done in a BSL-3 or higher-level laboratory.
- 24. D** HSV-2 infections produce genital lesions. Infants born prematurely with disseminated infection of HSV-2 from HSV-positive mothers have a mortality rate of 50% to 60%. Testing of pregnant women for antibody and cesarean section delivery can prevent most neonatal HSV infections because the virus enters the fetus during the delivery process.
- 25. B** Varicella-zoster virus is the cause of an infection with chickenpox. As an adult, the father is experiencing shingles, a reactivation of the virus. The virus lies dormant in the sensory (dorsal root) ganglia of the spinal nerves, and its reactivation produces a nonweeping, blisterlike rash on an inflamed skin base that follows the path of the underlying nerves.
- 26. B** Transmission of rabies (rhabdovirus) occurs primarily from the bite of a rabid animal, but in 20% of human rabies cases, there is no known exposure to rabid animals. Donor tissues are not routinely tested for rabies. Diagnosis is achieved by performing fluorescent antibody staining of the infected tissues and electron microscopy, which shows the presence of bullet-shaped virus particles.

27. While on a 7-day cruise to Vancouver and Alaska, a number of passengers reported to the ship's medical staff complaining of vomiting and diarrhea. Which is the *most likely* virus to have infected these mainly adult passengers?

- A.** Rotavirus
- B.** Parainfluenza virus
- C.** Respiratory syncytial virus
- D.** Norovirus (Norwalk or Norwalk-like viruses)

Microbiology/Select diagnosis/Viruses/2

28. SITUATION: A patient tested positive for HIV-1 infection with use of a RT-PCR

method. However, 1 week later, a second blood sample was collected and sent to a reference laboratory, which performed a confirmatory Western blot test. The Western blot test was negative. What *best* explains these results?

- A. The samples for RT-PCR and Western blot were not from the same patient
- B. RT-PCR is more sensitive for early detection
- C. The RT-PCR result was falsely positive
- D. The RT-PCR test detected HIV-2 and the Western blot is specific for HIV-1

Microbiology/Apply knowledge to identify sources of error/HIV testing/3

- 29. SITUATION A pregnant woman was seen by her obstetrician with signs of flu-like illness during her second trimester. NAT was ordered for *Toxoplasma gondii*, rubella, CMV, and HSV-1 and -2. All test results were negative. However, after delivery, the newborn exhibited signs of an infection and failed to survive. Which virus in the mother causes a 90% transplacental infection rate?**

- A. Hepatitis A
- B. Hepatitis B
- C. Hepatitis D
- D. Hepatitis G

Microbiology/Select diagnosis/Viruses/3

- 30. An immunocompromised patient was admitted to the hospital with a diagnosis of hemorrhagic cystitis. Which combination of virus and specimen would be *most* appropriate to diagnose a viral cause of this disorder?**

- A. BK virus—urine
- B. Human papilloma virus—skin
- C. Hepatitis B virus—serum
- D. Epstein—Barr virus—serum

Microbiology/Select identification/Viruses/2

Answers to Questions 27–30

- 27. D** Adult gastroenteritis caused by norovirus follows a 1 to 2 day incubation. Diagnosis is usually made on the basis of clinical criteria and confirmed by EIA, RT-PCR, or immune electron microscopy. Hepatitis E virus, also of the Caliciviridae family, also causes gastroenteritis 1 to 2 days after exposure but is not endemic in the United States and other developed countries.
- 28. B** Western blot is a test for detecting antibodies to HIV, and sufficient antibodies are not usually present in the first 3 to 4 weeks of infection to cause a positive test result. In some persons, antibodies do not appear until 3 months after the initial HIV infection. RT-PCR detects viral RNA and reduces the window phase of infection to approximately 1 week. RT-PCR tests for HIV-1 have 100% specificity at a cutoff of 20 to 40 copies RNA/mL. PCR is the preferred test for neonates because it avoids detection of maternal antibody and has replaced the Western blot as the confirmatory test of choice.
- 29. B** Tests for toxoplasmosis, German measles, CMV, and HSV comprise a panel of infectious agents known as the *TORCH panel*. The TORCH panel consists of serological tests for antibodies to these organisms and has largely been replaced by

more sensitive NAT when infection is suspected. The “O” in TORCH stands for “other,” and other viruses that cause transplacental infection, such as HBV and HIV are usually tested prenatally.

- 30. A** BK virus, a polyoma virus, is transmitted by direct contact with infected respiratory secretions. It has tropism for the urinary system and often causes a latent, asymptomatic infection in the kidneys. However, in immunocompromised patients, the virus is often implicated in renal and bladder infections. Kidney failure caused by BK virus is a significant concern in bone marrow and renal transplant recipients.

- 31. Which of the following genera and RNA viruses listed is the cause of viral hemorrhagic fever displaying many filamentous shapes, such as “U” or “6” and circular forms while being viewed under the electron microscope?**

A. Filovirus–Ebola virus
B. Flavivirus–West Nile virus
C. Picornavirus–poliovirus
D. Toga virus–western equine encephalitis virus

Microbiology/Select diagnosis/Viruses/2

- 32. Which of the following RNA flaviviruses is responsible for Guillain-Barre syndrome and birth defects of the brain (microcephaly, absent forehead, and triangular face shape) or poor outcomes in pregnant women and was first recognized in 2015 in Brazil?**

A. West Nile virus
B. Hepatitis C virus
C. Zika virus
D. Dengue fever virus

Microbiology/Select diagnosis/Viruses/2

- 33. Which of the following diseases, which are spread through direct contact and aerosols of the (RNA) coronavirus genera, was transmitted to hospital workers who were caring for infected patients in China, causing a severe acute respiratory illness that led to a worldwide outbreak in 2002?**

A. Severe acute respiratory syndrome (SARS)
B. Encephalitis
C. Gastroenteritis
D. Liver necrosis

Microbiology/Select diagnosis/Viruses/2

- 34. A pregnant patient presented with symptoms similar to hepatitis A but tested negative. She had traveled to India most recently. Which of the following is the cause of a hepatitis infection with a high mortality rate (10%–20%) in pregnant women and is transmitted via the fecal–oral route?**

A. Hepatitis B
B. Hepatitis E
C. Hepatitis C
D. Hepatitis D

Microbiology/Select diagnosis/Viruses/3

Answers to Questions 31–34.

- 31. A** The Ebola RNA virus is endemic in the Democratic Republic of the Congo with a human case fatality rate of 80% to 90%. The virus is transmitted from person to person or to caregivers through body fluids and respiratory droplets. Transmission of the virus to humans is from monkeys or other wild animals. It is speculated that bats harbor the virus as carriers or reservoirs. Using electron microscopy, the shape described is similar to that of the Marburg virus (“shepherd’s hook” shape), which causes a less severe viral hemorrhagic fever with a fatality rate of 23% to 25%. For diagnosis, electron microscopy (research laboratories), RT-PCR, and antibody detection with ELISA (IgG and IgM) are used along with cell culture in monkey cells in a BSL-4 laboratory. Both infections have no cure or established drug treatment. Several Ebola vaccines are being tested in clinical trials, as reported by the World Health Organization (WHO).
- 32. C** The Zika virus is transmitted by the bite of the *Aegypti* mosquito, and the virus disappears from blood after 2 weeks; however, the virus can linger in sperm for 2 months. Zika virus outbreaks were seen in Brazil and southern Mexico. According to the Pan-American Health Organization (PAHO), reports of local transmission of Zika virus occurred throughout Mexico, with a peak in number of infections occurring in October 2016. To curb the transmission of the virus, genetically modified mosquitoes were used; after mating, the offspring would die before reaching adulthood. Spraying with insecticides kills the adults and larvae, and this was one way used to control outbreaks that occurred in Miami in 2016.
- 33. A** SARS virus is detected with RT-PCR and can be cultured by using the Vero-E6 cell line. Low levels of the virus early in the infection poses a problem in diagnosing SARS.
- 34. B** Hepatitis E virus (HEV) infection is the result of an infection with a new genus, *Hepevirus*. It is transmitted by the fecal–oral route and mimics hepatitis A, except that HEV has a high rate of fatality (10%–20%) among pregnant women. It is not endemic in the United States or in other developed areas of the world. Pregnant women should be cautioned about drinking contaminated water and traveling to parts of the world where HEV is endemic (India and Pakistan). Fulminant hepatitis, with a fatality rate of 30%, results if infection occurs during the third trimester. Diagnosis is through serology (IgG and IgM antibodies) as well as liver function tests (aspartate aminotransferase [AST], alanine aminotransferase [ALT], and bilirubin) during the early phase of the infection.